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From Bias to Fairness: Can Synthetic Data Bridge the Gap in Tabular Machine Learning?

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Abstract. Synthetic data generation (SDG) has emerged as a promising strategy to address data scarcity and class imbalance in machine learning (ML), and has recently been explored for bias mitigation. However, its impact on fairness remains poorly understood, particularly in tabular domains where dependencies between protected attributes and target variables may be preserved or amplified by generative models. In this work, we present a systematic study of SDG for fairness in tabular classification. We evaluate SDG methods and strategies, as well as their combination with pre-processing bias mitigation techniques. Experiments are conducted across diverse real-world datasets, classifiers, and complementary performance and fairness metrics. Our results show that synthetic data is not inherently fairness-aware: depending on how it is applied, it can either mitigate or exacerbate bias, even when high data fidelity is achieved. We further characterize the fairness–performance trade-off, revealing a Pareto frontier in which improvements in fairness frequently come at the expense of utility. In particular, subgroup-targeted strategies consistently yield better fairness outcomes, while naive full-data augmentation often amplifies existing disparities. Moreover, a hybrid approach combining reweighing with subgroup-conditioned synthesis achieves strong fairness improvements with minimal loss in predictive performance, especially in high-bias scenarios. These findings provide practical guidance for the effective use of synthetic data in bias mitigation and highlight the limitations of current approaches when fairness is not explicitly incorporated into the data generation process.

Keywords: Synthetic data generation · Tabular data · Fairness in machine learning · Data fidelity · Bias mitigation.

1 Introduction

The integration of machine learning (ML) models into decision-making frameworks has been proliferating across high-stakes sectors, including healthcare, financial services, and criminal justice. However, their reliability and broader applicability can be compromised by biases present in the data, which may lead to systematically different outcomes across population groups [25]. These disparities can result in unequal access to opportunities, resources, or services, raising concerns about the trustworthiness and ethical implications of ML systems [13].

Empirical evidence shows that such biases often stem from imbalanced or non-representative data, as well as from structural inequalities reflected in real-world datasets, and may ultimately limit the safe deployment and generalization of these models [16,20].

In response, there has been a growing interest in fairness, defined here as the absence of bias, and in strategies to promote it throughout the ML pipeline [2,13]. Traditionally, bias mitigation methods are categorized into three groups: *pre-processing* (techniques applied to the training data), *in-processing* (strategies incorporated during model training), and *post-processing* (approaches applied to a trained model) [7]. Focusing on pre-processing strategies, which are directly related to this work, the most commonly adopted methods in tabular data include reweighting, resampling (e.g., over- and under-sampling), and data augmentation, all aimed at improving the representation of underrepresented groups and reducing disparities across protected attributes [7,20]. Additional techniques involve feature-level transformations, such as applying perturbations to reduce their influence, or leveraging data integration to construct more comprehensive representations of individuals [20].

Within this context, synthetic data generation (SDG) has emerged as a promising mechanism for bias mitigation, aiming to rebalance datasets or generate more representative samples of underrepresented groups [16]. Previously, the use of synthethic data has been common for both data augmentation and privacy-preserving data sharing, with a wide range of approaches proposed, from classical oversampling techniques to advanced generative models [11]. However, synthetic data is not inherently fairness-aware, as generative models tend to replicate the statistical properties of the original data, including existing biases [21]. Thus, theoretically, they may propagate or even amplify bias under certain conditions. Furthermore, while synthetic data can implicitly influence learning dynamics, its interaction with fairness objectives remains insufficiently understood, particularly in tabular domains where complex dependencies between protected attributes and target variables may be preserved or distorted.

In this work, we present a systematic study on the role of synthetic data for bias mitigation in tabular ML. We investigate multiple types of synthetic data generators and distinct data generation strategies, including full-data augmentation and subgroup-based approaches, as well as their interaction with established pre-processing bias mitigation techniques. Our evaluation considers diverse datasets, predictive models, and complementary definitions of fairness, enabling a comprehensive analysis of fairness–performance trade-offs. Our findings highlight the importance of how synthetic data is generated and applied, and provide practical insights into when and how it can be effectively used for bias mitigation in tabular settings.

2 Related Work

Recent studies have increasingly investigated the role of SDGs in addressing bias and fairness in ML, particularly in the presence of class and group imbalance

[21]. A recurrent finding across this literature is that data imbalance, both in target classes and protected groups, is a key driver of unfair predictions, as models tend to underperform on underrepresented populations [16]. Synthetic data has therefore been explored as a mean to mitigate imbalance while also addressing privacy constraints, especially in sensitive domains such as healthcare and education.

To address data-level bias (also referred to as *pre-training bias*), a variety of SDG methods have been proposed. Early approaches rely on data augmentation techniques such as SMOTE and its variants [21], as well as statistical methods like Gaussian Copula, which model dependencies through marginal distributions and a copula function [15]. More recent work leverages advanced generative models to produce high-fidelity tabular data, including GAN- and VAE-based approaches such as CTGAN and TVAE [24], as well as diffusion-based models [9]. Additionally, some methods explicitly incorporate fairness constraints into the generation process, such as adversarial frameworks (e.g., TabFairGAN [18]) and causally-aware models (e.g., DECAF [22]). However, incorporating fairness constraints during generation does not necessarily translate into improved predictive performance or consistent fairness gains in downstream tasks.

Overall, findings from comparative studies [11,16] suggest that synthetic data can act as a data-centric intervention to improve fairness, particularly when targeting underrepresented subgroups. However, these benefits are not consistent: the effectiveness of synthetic data depends strongly on the quality of the generator and the characteristics of the original dataset. Moreover, important trade-offs have been identified between fairness, utility, and privacy, with no single approach consistently optimizing all three [11]. In addition, synthetic data tends to inherit the statistical properties of the original data, including existing biases, which raises concerns about its reliability as a standalone fairness intervention [23]. Recent evidence also indicates that using pre-processing mitigation methods to debias synthetic data can lead to improved fairness outcomes compared to using SDGs alone [11], although this interaction remains underexplored.

Despite these advances, existing work typically evaluates synthetic data generation in isolation or under limited experimental settings, often focusing on a single generation strategy or a fixed application pipeline. In particular, there is a lack of systematic studies comparing multiple types of synthetic data generators and different data generation strategies (e.g., full-data augmentation versus targeted subgroup generation), as well as their interaction with established pre-processing bias mitigation methods. Moreover, prior work often relies on a limited set of fairness metrics, without considering how different definitions of fairness may lead to divergent conclusions. Addressing this gap is essential to better understand how different design choices influence fairness–performance trade-offs in tabular ML. In this work, we aim to provide such a systematic evaluation by jointly analyzing multiple SDGs, diverse data generation schemes, their combination with pre-processing techniques, and multiple manifestations of fairness.

Table 1: Summary of the datasets used in this study after preprocessing.

Dataset	#Instances	#Attributes	Target	Protected Attribute	Favorable Class
COMPAS	5,278	11	two_year_recid	race	No recidivism (0)
Adult Income	10,000	55	income	sex	>50K (1)
Diabetes	10,000	40	readmitted	race	No readmission (0)

3 Materials and Methods

This section describes the datasets used in our experiments, as well as the bias mitigation methods, including traditional in-processing methods and syntethic data generation approaches. Moreover, we define the evaluation metrics used for fairness and utility. Code and experimental results are available at https://github.com/mlab-inf-ufrgs/bias2fair-synth.

3.1 Dataset collection and preparation

We evaluate our study on binary classification tasks based on three widely used tabular datasets from distinct application domains: criminal justice (COMPAS), socio-economic prediction (Adult Income), and healthcare (Diabetes 130-US Hospitals). The characteristics of the final versions of datasets, after preprocessing steps, are summarized in Table 1.

Target and protected attributes. The COMPAS dataset (n = 5,728) focuses on recidivism prediction, with race as the protected attribute and non-recidivism as the favorable outcome [17]. The Adult dataset (n = 48,842) models income prediction, using sex as the protected attribute and high income ($>50K$) as the favorable class [1]. The Diabetes (n = 95,309) dataset addresses hospital readmission prediction, considering race as the protected attribute and no readmission as the favorable outcome [3].

Preprocessing steps. Problematic features were removed across all datasets, including those with high missingness, identifiers or administrative attributes, and non-predictive or noisy variables (e.g., IDs, textual fields, and high-cardinality features), as well as features that could introduce confounding effects with the sensitive variable or potential leakage sources. Protected attributes were binarized following commonly adopted fairness conventions, where disadvantaged groups were mapped to 0 and privileged groups to 1 (e.g., African-American = 0 and Caucasian = 1 for race; Female = 0 and Male = 1 for gender). Categorical features were transformed using one-hot encoding and ordinal mapping, depending on their nature. Numerical attributes were standardized using z-score normalization. In the Diabetes dataset, additional transformations were applied to age groups, laboratory results, and medication-related variables. Finally, due to computational costs, the Adult and Diabetes datasets were sampled to 10,000 instances, respecting the original relation between protected attribute and target class.

3.2 Synthetic data generation methods

We consider SDG methods from different modeling strategies, encompassing probabilistic, adversarial, and diffusion-based approaches. TVAE (Tabular Variational Autoencoder) is a latent variable model based on variational autoencoders that learns a compressed representation of tabular data to generate new samples [24]. CTGAN (Conditional Tabular GAN) is an adversarial model specifically designed for tabular data, which leverages conditional generation to handle imbalanced categorical variables and improve sample diversity [24]. In contrast, the Gaussian Copula represents a classical statistical approach, modeling dependencies between variables through marginal distributions combined with a Gaussian copula [15]. Finally, Tabular Diffusion Models are based on stochastic diffusion processes, where noise is iteratively added and removed to learn the data distribution and generate realistic synthetic samples [9].

3.3 Pre-processing bias mitigation methods

We also consider representative pre-processing bias mitigation techniques that operate directly on the training data before model learning, thus being a good baseline for our experiments [7,20]. Reweighting adjusts the importance of instances based on the joint distribution of the target and the protected attribute, aiming to balance group representation and reduce disparities [8]. The Disparate Impact Remover (DIR) modifies feature values through a transformation that reduces their dependence on the protected attribute while preserving as much predictive information as possible [4]. Finally, Learning Fair Representations (LFR) learns a latent representation of the data that simultaneously seeks to retain task-relevant information and obfuscate protected attributes, producing a transformed dataset that promotes fairer downstream predictions [25].

3.4 Model training and evaluation

Datasets were split into training and test sets (80/20) using stratified sampling based on both the target and the protected attribute. Experiments were repeated with 10 random seeds, and results are reported as mean and standard deviation. We evaluated three supervised learning algorithms: Logistic Regression (max_iter=1000), CatBoost ($iterations$=500), and Support Vector Machines (SVM, with probability estimation enabled). Default hyperparameters were used to isolate the effects of data-centric interventions and ensure comparability across settings.

3.5 Evaluation metrics

We evaluate the impact of synthetic data along three complementary dimensions: data fidelity, ML utility, and fairness.

Data Fidelity Data fidelity is assessed using the Jensen–Shannon divergence to measure the similarity between real and synthetic data distributions, complemented by visual comparisons of key variable distributions [14].

ML Utility The utility of synthetic data is evaluated in terms of its contribution to downstream classification performance when combined with real data during training. Specifically, we assess whether the inclusion of synthetic samples improves or degrades model performance, using F1-score as the primary metric. Additional metrics are reported in the online repository.

Fairness Fairness is evaluated using group-based metrics that capture disparities in predictions across groups defined by protected attributes. Since different fairness definitions reflect distinct notions of equity. Considering that $\hat{Y}$ denotes the model prediction, Y the true label, and A the protected attribute [5,4], the following metrics were selected:

Disparate Impact (DI): measures the ratio of favorable outcomes between groups, capturing disparities in overall prediction rates. Values close to 1 indicate parity, while deviations suggest potential bias.

$$\text{DI} = \frac{P(\hat{Y} = 1 \mid A = 0)}{P(\hat{Y} = 1 \mid A = 1)} \tag{1}$$

Equal Opportunity Difference (EOD): measures the difference in true positive rates between groups. A value close to 0 indicates similar opportunity across groups. This metric is aligned with *separation-based* fairness.

$$\text{EOD} = P(\hat{Y} = 1 \mid Y = 1, A = 0) - P(\hat{Y} = 1 \mid Y = 1, A = 1) \tag{2}$$

Average Absolute Odds Difference (AAOD): computes the average of true positive and false positive rates, providing a more comprehensive measure of error-based fairness. Lower values indicate more balanced error rates across groups.

$$\begin{aligned} \text{AAOD} = \frac{1}{2}\Big(&\Big|P(\hat{Y} = 1 \mid Y = 1, A = 0) - P(\hat{Y} = 1 \mid Y = 1, A = 1)\Big| \\ &+ \Big|P(\hat{Y} = 1 \mid Y = 0, A = 0) - P(\hat{Y} = 1 \mid Y = 0, A = 1)\Big| \Big) \end{aligned} \tag{3}$$

To assess whether the observed differences are statistically meaningful, we conducted paired Wilcoxon signed-rank tests across all experimental configurations, considering results statistically significant at $p < 0.05$.

4 Experimental Design and Setup

To systematically evaluate the impact of synthetic data on fairness, we define six experimental scenarios, representing distinct intervention strategies:

S1: Original (baseline)
Predictive models are trained and evaluated using the original data, without any bias mitigation strategy (neither with pre-processing mitigation methods or SDGs). This scenario serves as the baseline for comparison.

S2: Pre-processing mitigation method
Three classical bias mitigation techniques are applied at the data preprocessing stage: Reweighing, DIR, and LFR.

S3: Full synthetic data augmentation
Synthetic data is generated from the full training dataset without conditioning or explicit bias mitigation objectives. This scenario acts as a control to isolate the effect of data augmentation. The amount of synthetic data is controlled by a scaling factor $\alpha \in \{1.5, 2, 3\}$, where $\alpha = 1.5$ increases the dataset size by 50%, $\alpha = 2$ doubles it, and $\alpha = 3$ triples it.

S4: Synthetic oversampling of minority groups
Synthetic data are generated specifically for underrepresented groups, guided by the distribution of majority groups, with the goal of balancing the joint distribution between the protected attribute and the target variable. This strategy aims to improve representation without modifying the majority group. Let A denote the protected attribute and Y the target variable, with counts organized in a contingency matrix. The maximum observed value across all combinations of A and Y is taken as the reference. For each pair (A, Y), a number of synthetic samples is generated equal to the difference between this maximum value and the observed count, so that all cells reach the same frequency.

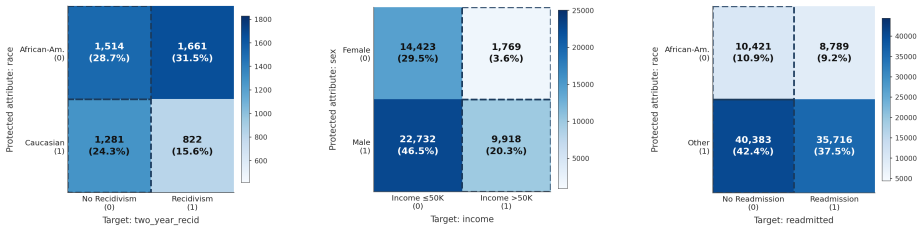
S5: Subgroup-based SDG (Target $\times$ Protected Attribute)
Synthetic data generation is performed at the subgroup level, considering the intersection between the protected attribute and the target variable. Unlike S4, the balancing here is guided by a global average value. An ideal value N_{ideal} is defined for each cell of the contingency matrix (protected attribute $\times$ target), computed as the ratio between the total number of instances in the dataset and the number of possible combinations between these variables, approximating demographic parity. Next, for each cell, the difference between N_{ideal} and the observed frequency is computed. This difference determines the number of synthetic samples to be added (for cells below the ideal value), so that all combinations reach approximately the same size. In this way, the method promotes balance across subgroups, reducing imbalance within each protected $\times$ outcome stratum without removing any original data.

S6: Hybrid mitigation (pre-processing + synthetic data)
This scenario combines Reweighing (from S2) with subgroup-based synthetic generation (S5), allowing us to assess potential synergistic or complementary effects between these two approaches.

All experiments were conducted on a machine equipped with an AMD Ryzen 7 7700X CPU (8 cores, 16 threads) and an NVIDIA GeForce RTX 5080 GPU with 16 GB of VRAM. The implementation was carried out in Python 3.12, using standard libraries for data processing, modeling, and fairness evaluation, including *NumPy*, *Pandas*, *Scikit-learn*, *Synthetic Data Vault (SDV)*, *Fairlearn*, *AIF360*, and *CatBoost*.

5 Results

In the following sections, we first validate the quality of synthetic data, and then present dataset-specific analyses, followed by a cross-dataset comparison and statistical significance assessment. Figure 1 presents the joint distribution for protected attribute (rows) and target attribute (columns) for each dataset.



(a) COMPAS (b) Adult (c) Diabetes

Fig. 1: Contingency matrices (protected attribute × target variable) for the COMPAS, Adult, and Diabetes datasets.

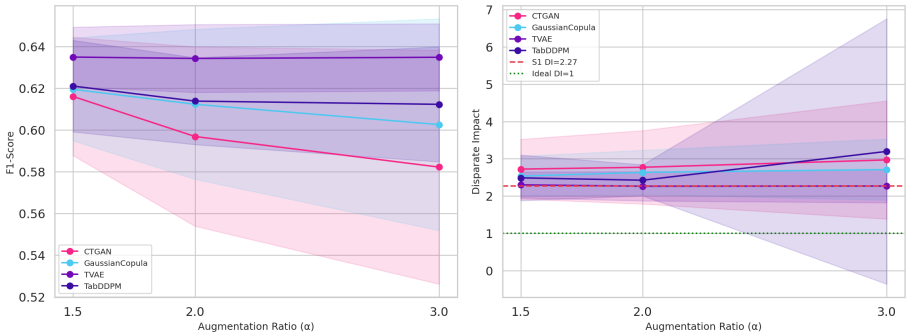
5.1 Synthetic fidelity analysis

To ensure that performance differences are not driven by poor data quality, we evaluate synthetic fidelity using Jensen–Shannon Divergence (JSD) between real and synthetic feature distributions. Overall, all generators achieve acceptable fidelity (JSD < 0.15) across most features, with TVAE and Gaussian Copula consistently yielding the lowest divergence. The analysis includes protected attributes, target variables, and key numerical features. Detailed plots are available in the project repository.

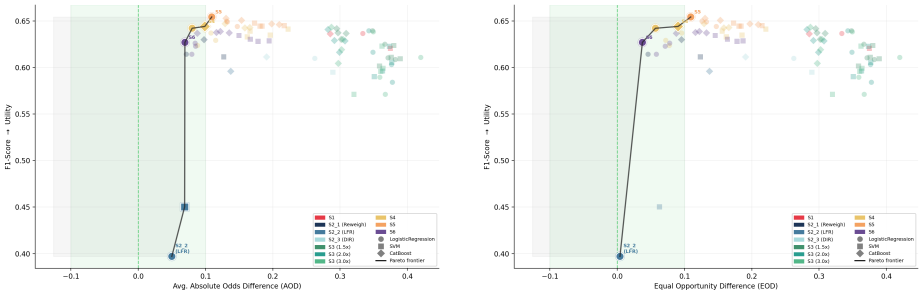
5.2 COMPAS: racial bias in recidivism prediction

The COMPAS dataset is characterized by a high level of pre-existing racial bias (DI ≈ 2.27 in S1),where African-American defendants are substantially more likely to receive a positive (high-risk) prediction. Table 2 presents a comparative analysis between ML utility (as measured by F1) and fairness across different bias mitigation scenarios. Pre-processing techniques (S2), such as LFR and Reweighing, are able to drastically reduce disparity (i.e., DI closer to 1), but at the cost of a notable drop in F1 performance, especially for LFR (0.481).

In contrast, the hybrid scenarios (S4, S5, and S6) demonstrate the best balance: they not only maintain or surpass the baseline utility (0.631), with S5 reaching 0.648, but also promote significant fairness improvements (a reduction of ΔDI of approximately -0.9). The purely synthetic approach (S3), however,



(a) Effect of synthetic augmentation ratio (α) in scenario S3



(b) AAOD (c) EOD

Fig. 2: Effect of synthetic data on fairness and performance for the COMPAS dataset: augmentation sensitivity (top) and trade-offs with Pareto frontiers (bottom).

fails to mitigate bias, resulting in increased disparity (positive ΔDI) relative to the original baseline, effectively amplifying pre-existing bias, as shown in Fig. 2a. Figures 2b and 2c further illustrates that scenario S2 achieves the strongest fairness metrics overall. Nevertheless, scenario S6 provides the best trade-off between utility and fairness, representing the most balanced configuration, followed by scenario S5.

5.3 Adult Income: gender bias in income prediction

The Adult dataset exhibits moderate systematic bias against women in income prediction (DI $\approx$ 0.292 in S1).Here, DI < 1 indicates that the privileged group (males) receives positive predictions at a higher rate. Table 3 shows that purely synthetic augmentation (S3) leads to only marginal changes, with slight degradations in DI and a gradual decline in performance as the *augmentation* factor increases. In contrast, pre-processing methods (S2) yield mixed results: LFR reduces disparity but severely degrades performance (F1 $\approx$ 0.305), whereas Reweighing provides a better *trade-off*, improving fairness (DI $\approx$ 0.619) while

Table 2: Utility (F1) and fairness (DI, AOD, EOD) by scenario (± standard deviation) for the COMPAS dataset. ΔDI: change in DI from baseline (S1); negative = improved fairness, positive = increased disparity.

Scenario	**F1**	**DI**	**ΔDI**	**AOD**	**EOD**
S1 (Baseline)	0.631 ± 0.014	2.268 ± 0.378	0.000	0.331 ± 0.061	0.331 ± 0.061
S2 Pre-processing					
DIRemover	0.605 ± 0.026	1.996 ± 0.373	-0.272	0.248 ± 0.077	0.248 ± 0.077
LFR	0.481 ± 0.182	1.257 ± 0.282	-1.010	0.086 ± 0.064	0.068 ± 0.078
Reweighing	0.618 ± 0.014	1.225 ± 0.141	-1.042	0.099 ± 0.050	0.088 ± 0.061
S3 Synthetic					
$\alpha = 1.5$	0.623 ± 0.024	2.511 ± 0.607	+0.243	0.348 ± 0.068	0.348 ± 0.068
$\alpha = 2.0$	0.614 ± 0.033	2.523 ± 0.663	+0.255	0.341 ± 0.065	0.341 ± 0.065
$\alpha = 3.0$	0.608 ± 0.045	2.786 ± 2.007	+0.518	0.341 ± 0.079	0.341 ± 0.079
Synthetic Conditional					
S4	0.638 ± 0.014	1.326 ± 0.236	-0.941	0.141 ± 0.074	0.131 ± 0.086
S5	0.648 ± 0.013	1.397 ± 0.172	-0.871	0.164 ± 0.064	0.163 ± 0.065
S6	0.630 ± 0.014	1.301 ± 0.204	-0.967	0.123 ± 0.065	0.115 ± 0.073

maintaining competitive performance. Hybrid strategies (S4–S6) achieve the most balanced outcomes, with S5 and S6 improving DI (up to ≈ 0.450) while preserving performance close to the *baseline* (F1 ≈ 0.65). This indicates that targeted synthetic data, when combined with mitigation strategies, can reduce disparities without substantial loss of utility. Overall, the results reinforce that synthetic data is not inherently fairness-aware and is more effective when combined with explicit bias mitigation strategies.

5.4 Diabetes: race bias in readmission prediction

The Diabetes dataset presents a distinct fairness profile, with low initial bias (DI ≈ 0.913 in S1), making it a challenging setting in which interventions may risk introducing new disparities. Table 4 shows that this low-asymmetry scenario makes the dataset relatively resistant to bias mitigation strategies. Preprocessing methods such as *DIRemover* and *Reweighing* maintain performance close to the baseline (F1 ≈ 0.50–0.52), but do not provide consistent improvements in fairness. In contrast, the *LFR* method significantly degrades model utility (F1 ≈ 0.245), despite bringing DI closer to more desirable values. The most interesting pattern emerges in the hybrid strategies. Scenario S4 improves F1 (≈ 0.537), but introduces greater instability in fairness. Scenario S5 (*fair synthetic*) worsens fairness consistency (DI > 1), suggesting that targeted synthetic data generation may introduce new biases in datasets that are originally only weakly biased. In contrast, S6 provides the best overall balance, maintaining competitive performance (F1 ≈ 0.52) while better controlling fairness drift.

Table 3: Utility (F1) and fairness (DI, AOD, EOD) by scenario (± standard deviation) for the Adult dataset. ΔDI: change in DI from baseline (S1); negative = increased disparity, positive = improved fairness.

Scenario	**F1**	**DI**	**ΔDI**	**AOD**	**EOD**
S1 (Baseline)	0.655 ± 0.032	0.292 ± 0.024	0.000	0.118 ± 0.036	-0.116 ± 0.040
S2 Pre-processing					
DIRemover	0.626 ± 0.013	0.208 ± 0.068	-0.084	0.221 ± 0.110	-0.219 ± 0.113
LFR	0.305 ± 0.025	0.386 ± 0.087	+0.094	0.032 ± 0.020	-0.000 ± 0.038
Reweighing	0.631 ± 0.035	0.619 ± 0.062	+0.326	0.142 ± 0.038	0.142 ± 0.038
S3 Synthetic					
$\alpha = 1.5$	0.627 ± 0.047	0.273 ± 0.031	-0.020	0.140 ± 0.040	-0.138 ± 0.044
$\alpha = 2.0$	0.613 ± 0.058	0.266 ± 0.039	-0.026	0.147 ± 0.044	-0.146 ± 0.047
$\alpha = 3.0$	0.593 ± 0.082	0.257 ± 0.048	-0.035	0.148 ± 0.053	-0.147 ± 0.055
Synthetic Conditional					
S4	0.639 ± 0.040	0.371 ± 0.098	+0.079	0.133 ± 0.064	-0.090 ± 0.094
S5	0.648 ± 0.034	0.450 ± 0.134	+0.158	0.089 ± 0.037	-0.014 ± 0.093
S6	0.654 ± 0.033	0.396 ± 0.102	+0.104	0.099 ± 0.041	-0.054 ± 0.084

5.5 Cross-dataset comparison of disparate impact

The three datasets present fundamentally different initial bias profiles as measured by DI, which directly shapes both the type and magnitude of fairness gains achievable through the evaluated interventions. Table 5 summarizes the baseline DI values (S1) and the best DI achieved across all scenarios for each dataset. COMPAS shows the largest initial disparity (DI ≈ 2.27), with African-American defendants receiving favorable outcomes at over twice the rate of Caucasians. This allows substantial reductions: pre-processing methods (Reweighing, LFR) lower DI by more than 1.0, while hybrid approaches (S6) achieve similar gains (ΔDI ≈ -0.97) with better utility. In contrast, Adult presents an inverted disparity (DI ≈ 0.29), where women receive favorable predictions at about one-third the rate of men. Although ΔDI is smaller, the gap to parity is large; Reweighing yields the biggest improvement (ΔDI $= +0.33$), while hybrid methods offer moderate, utility-preserving gains. Diabetes starts near parity (DI ≈ 0.91), limiting improvements without introducing bias. All methods yield $|\Delta\text{DI}| \leq 0.11$, and some (LFR, S5) overshoot (DI > 1), reversing disparity. Overall, initial bias severity and direction determine the best strategy: strong interventions suit high-bias cases (COMPAS, Adult), while conservative methods (e.g., Reweighing) are safer in near-fair settings (Diabetes).

5.6 Statistical Significance Analysis: Wilcoxon Tests

The statistical analyses confirm that hybrid strategies S4 and S5 yield statistically significant improvements in DI on the COMPAS and Adult datasets ($p < 0.05$). In contrast, the purely synthetic approach (S3) does not produce statistically significant improvements in DI in any configuration. Additionally,

Table 4: Utility (F1) and fairness (DI, AOD, EOD) by scenario (± standard deviation) for the Diabetes dataset. ΔDI: change in DI from baseline (S1); negative = increased disparity, positive = improved fairness.

Scenario	**F1**	**DI**	**ΔDI**	**AOD**	**EOD**
S1 (Baseline)	0.516 ± 0.029	0.913 ± 0.068	0.000	0.050 ± 0.022	-0.025 ± 0.035
S2 Pre-processing					
DIRemover	0.505 ± 0.037	0.992 ± 0.216	+0.078	0.081 ± 0.044	-0.004 ± 0.080
LFR	0.245 ± 0.173	1.006 ± 0.462	+0.092	0.033 ± 0.028	0.015 ± 0.032
Reweighing	0.516 ± 0.029	0.961 ± 0.072	+0.047	0.043 ± 0.023	-0.007 ± 0.040
S3 Synthetic					
$\alpha = 1.5$	0.504 ± 0.050	0.911 ± 0.128	-0.002	0.063 ± 0.038	-0.026 ± 0.053
$\alpha = 2.0$	0.496 ± 0.066	0.912 ± 0.166	-0.002	0.071 ± 0.048	-0.027 ± 0.060
$\alpha = 3.0$	0.488 ± 0.089	0.935 ± 0.244	+0.022	0.079 ± 0.057	-0.024 ± 0.073
Synthetic Conditional					
S4	0.537 ± 0.031	0.838 ± 0.274	-0.075	0.123 ± 0.077	-0.073 ± 0.105
S5	0.515 ± 0.034	1.019 ± 0.252	+0.106	0.086 ± 0.059	0.007 ± 0.091
S6	0.520 ± 0.035	0.888 ± 0.238	-0.025	0.086 ± 0.060	-0.045 ± 0.084

Table 5: Cross-dataset DI comparison: baseline, best fairness, and best utility-fairness trade-off.

			Best Fairness			**Best Trade-off**		
Dataset	**Profile**	**DI_{S1}**	**Scenario**	**DI**	**ΔDI**	**Scenario**	**F1**	**DI**
COMPAS	High (>1)	2.268	Reweighing	1.225	−1.042	S6	0.630	1.301
Adult	High (<1)	0.292	Reweighing	0.619	+0.326	S6	0.654	0.396
Diabetes	Low (≈1)	0.913	Reweighing	0.961	+0.047	S6	0.520	0.888

the LFR method consistently leads to significant degradation in predictive performance (F1) across all datasets ($p < 0.01$).

6 Discussion

Our results provide evidence of a proportional amplification effect: SDGs that preserve the original joint distribution also preserve and, in many cases, amplify existing disparities between groups. This reflects a structural property rather than a limitation of any specific generator. Simply augmenting training data with high-fidelity synthetic samples, without conditional controls, should not be expected to yield improvements in fairness, as also demonstrated by [23].

S4 and S5 are effective precisely because they break the proportionality effect by explicitly conditioning the data generation process on underrepresented subgroups. In S4, samples from the minority group are generated based on the distribution of the majority group, effectively reducing the representation gap between groups. In S5, conditioning on (sensitive attribute $\times$ outcome) corrects imbalances within specific subgroups that traditional oversampling methods fail to address [6] [12]. This approach is grounded in the causal theory of fairness [10]: by intervening in the data generation process at the level of specific strata (A, Y), rather than acting directly on the prediction function, the causal influence of A on $\hat{Y}$ is reduced. It can simultaneously improve both fairness and utility, as observed in S4 for the COMPAS dataset. Thus, bias mitigation does not necessarily lead to a degradation in predictive performance across all conditions [19].

The Diabetes dataset demonstrates that fairness interventions should be calibrated to the initial level of bias. In near-parity scenarios, aggressive interventions (such as LFR or S3 with high α) may introduce new disparities. Reweighing emerges as the most robust method in this regime, achieving near-zero statistical parity difference (SPD) without sacrificing utility. These findings suggest that the initial level of bias should be assessed before selecting an appropriate intervention strategy.

7 Limitations and Future Work

This study is limited to the evaluation of binary protected attributes and does not address intersectionality across multiple protected attributes. Additionally, the Adult and Diabetes datasets were restricted to 10,000 instances due to computational constraints; therefore, experiments conducted on the full datasets may reveal different scalability behaviors. Finally, the analysis focuses on classification tasks, leaving fairness considerations in regression settings unexplored.

As future work, we propose extending the approach to include in-processing and post-processing stages, incorporating synthetic data augmentation techniques. Furthermore, we recommend investigating fairness in scenarios involving multiple protected attributes (e.g., race $\times$ gender). Lastly, we highlight the importance of adopting causal models for conditional data synthesis, aiming to generate counterfactually fair datasets.

8 Conclusion

This work investigated the use of synthetic data as a pre-processing strategy to mitigate bias in classification tasks. Across six experimental scenarios, ranging from a baseline without intervention to hybrid approaches, the results reveal a consistent pattern: generators that accurately reproduce the original data distribution also reproduce, and sometimes amplify, existing inequalities. Untargeted augmentation (S3) did not reduce bias and often increased disparities, reflecting a proportional amplification effect inherent to high-fidelity generation.

In contrast, targeted strategies significantly improved fairness. Minority group oversampling (S4) and subgroup-conditioned generation (S5) effectively broke this effect by intervening in underrepresented strata, without sacrificing performance, and in some cases even improving it. The hybrid approach (S6), combining Reweighing with subgroup-based generation, achieved the best overall balance between fairness and utility in the Pareto frontier.

In the Adult dataset, S4 and S5 significantly improved Disparate Impact, whereas LFR led to substantial performance degradation. In the Diabetes dataset, which exhibits low initial bias, more aggressive interventions introduced new imbalances; in this setting, Reweighing proved to be the most robust strategy.

In summary: (1) synthetic data is not a universal solution, and its effectiveness depends on conditional generation; (2) the most effective strategies are guided by causal principles and focus on subgroup-level interventions; and (3) hybrid approaches outperform standalone methods. As limitations, this study considered only binary protected attributes and binary classification tasks, and relied on subsets of the original datasets. Future work should explore intersectional fairness, other task types, and more advanced causal methods.

References

1. Becker, B., Kohavi, R.: Adult (1996). https://doi.org/10.24432/C5XW20
2. Caton, S., Haas, C.: Fairness in machine learning: A survey. ACM Computing Surveys **56**(7) (Apr 2024)
3. Clore, J., Cios, K., DeShazo, J., Strack, B.: Diabetes 130-us hospitals for years 1999-2008 (2014). https://doi.org/10.24432/C5230J
4. Feldman, M., Friedler, S.A., Moeller, J., Scheidegger, C., Venkatasubramanian, S.: Certifying and removing disparate impact. In: Proceedings of the 21th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. pp. 259–268 (2015)
5. Hardt, M., Price, E., Srebro, N.: Equality of opportunity in supervised learning. Advances in Neural Information Processing Systems **29** (2016)
6. Hastings Blow, C., Qian, L., Gibson, C., Obiomon, P., Dong, X.: Data augmentation via diffusion model to enhance ai fairness. Frontiers in Artificial Intelligence **8**, 1530397 (2025)
7. Hort, M., Chen, Z., Zhang, J.M., Harman, M., Sarro, F.: Bias mitigation for machine learning classifiers: A comprehensive survey. ACM Journal on Responsible Computing **1**(2), 1–52 (2024)

8. Kamiran, F., Calders, T.: Data preprocessing techniques for classification without discrimination. Knowledge and Information Systems **33**(1), 1–33 (2012)
9. Kotelnikov, A., Baranchuk, D., Rubachev, I., Babenko, A.: Tabddpm: Modelling tabular data with diffusion models. In: International Conference on Machine Learning. pp. 17564–17579. PMLR (2023)
10. Kusner, M.J., Loftus, J., Russell, C., Silva, R.: Counterfactual fairness. Advances in Neural Information Processing Systems **30** (2017)
11. Liu, Q., Deho, O., Vadiee, F., Khalil, M., Joksimovic, S., Siemens, G.: Can synthetic data be fair and private? a comparative study of synthetic data generation and fairness algorithms. In: Proceedings of the 15th International Learning Analytics and Knowledge Conference. pp. 591–600 (2025)
12. Marchesi, R., Micheletti, N., I-Hsien Kuo, N., Barbieri, S., Jurman, G., Osmani, V.: Generative ai mitigates representation bias and improves model fairness through synthetic health data. PLOS Computational Biology **21**(5), e1013080 (2025)
13. Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K., Galstyan, A.: A survey on bias and fairness in machine learning. ACM Computing Surveys **54**(6) (Jul 2021)
14. Nalewajski, R.F.: Elements of information theory. In: Perspectives in Electronic Structure Theory, pp. 371–395. Springer (2011)
15. Nelsen, R.B.: An introduction to copulas. Springer (2006)
16. Panagiotou, E., Roy, A., Ntoutsi, E.: Synthetic tabular data generation for class imbalance and fairness: A comparative study. arXiv preprint arXiv:2409.05215 (2024)
17. ProPublica: Compas analysis. `https://github.com/propublica/compas-analysis` (2016), accessed on: Apr. 29, 2026
18. Rajabi, A., Garibay, O.O.: Tabfairgan: Fair tabular data generation with generative adversarial networks. Machine Learning and Knowledge Extraction **4**(2), 488–501 (2022)
19. Rodolfa, K.T., Lamba, H., Ghani, R.: Empirical observation of negligible fairness–accuracy trade-offs in machine learning for public policy. Nature Machine Intelligence **3**(10), 896–904 (2021)
20. Ruback, L., Fortunato, R., Renso, C., Duarte, L.T., Teles, A.S.: Bias mitigation in machine learning models trained on structured health data: A systematic review. IEEE Access **14**, 46331–46348 (2026)
21. Shahul Hameed, M.A., Qureshi, A.M., Kaushik, A.: Bias mitigation via synthetic data generation: a review. Electronics **13**(19), 3909 (2024)
22. Van Breugel, B., Kyono, T., Berrevoets, J., Van der Schaar, M.: Decaf: Generating fair synthetic data using causally-aware generative networks. Advances in Neural Information Processing Systems **34**, 22221–22233 (2021)
23. Wyllie, S., Shumailov, I., Papernot, N.: Fairness feedback loops: training on synthetic data amplifies bias. In: Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency. pp. 2113–2147 (2024)
24. Xu, L., Skoularidou, M., Cuesta-Infante, A., Veeramachaneni, K.: Modeling tabular data using conditional GAN. Advances in Neural Information Processing Systems **32** (2019)
25. Zemel, R., Wu, Y., Swersky, K., Pitassi, T., Dwork, C.: Learning fair representations. In: International Conference on Machine Learning. pp. 325–333. PMLR (2013)